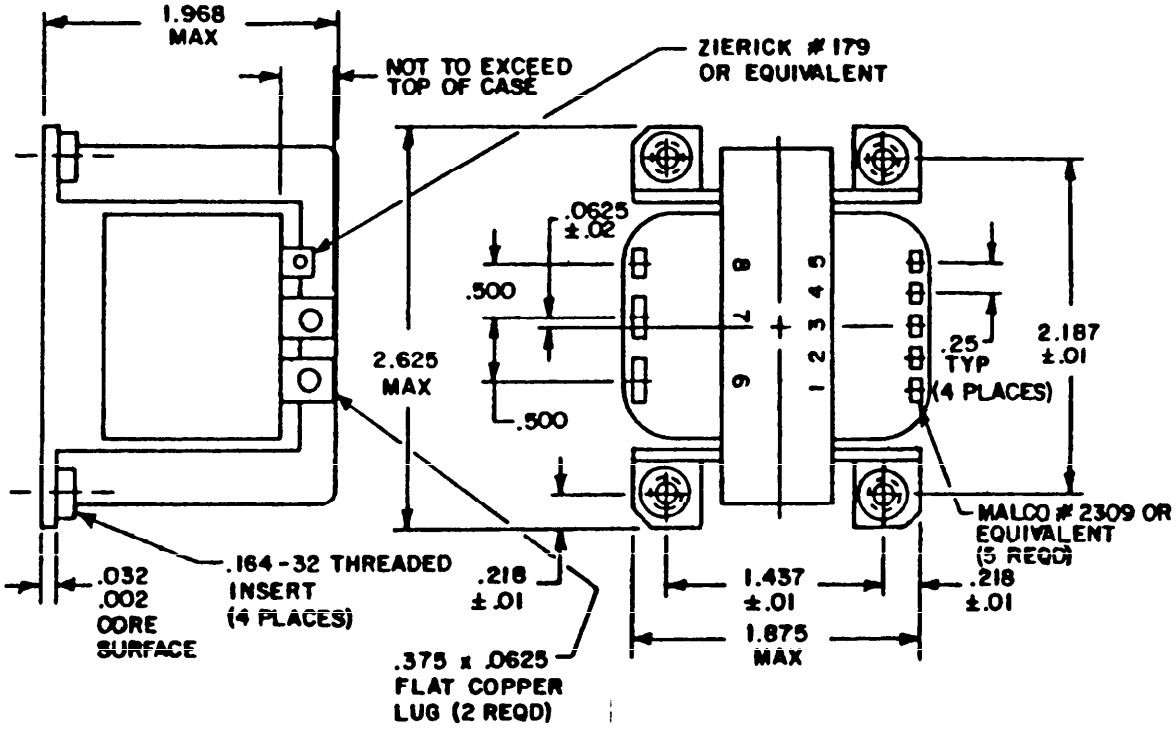
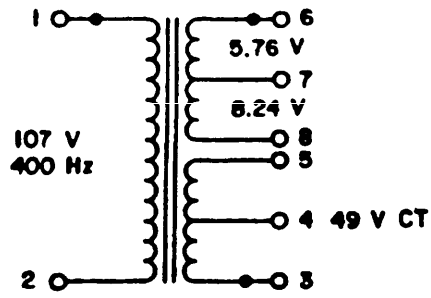
MILITARY SPECIFICATION SHEET

# TRANSFORMER, POWER, STEP-DOWN, TF5T03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-T-27.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .01 | 0.3 |
| .02 | 0.5 |
| .032 | 0.81 |
| .0625 | 1.588 |
| .164 | 4.17 |
| .218 | 5.54 |
| .25 | 6.4 |
| .305 | 7.75 |
| .500 | 12.70 |
| 1.437 | 36.50 |
| 1.875 | 47.62 |
| 1.968 | 49.99 |
| 2.187 | 55.55 |
| 2.625 | 66.68 |

WORKING VOLTAGE: 120 Vrms
ALTITUDE: 70 k FT MAX

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on side of case.

FIGURE 1. Dimensions and configuration.

(A) denotes changes


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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 107 volts rms, 400 hertz ±20.

Primary current (1-2): 2 ampere.

Voltampere rating: 214 voltamperes.

Secondary voltage:
(3-5): 49 volts ct.
(6-7): 5.76 volts.
(7-8): 8.24 volts.

Secondary current:
(3-5): 0.010 ampere.
(6-7): 30 amperes.
(7-8): 2.14 amperes.

Working voltage: 120 volts rms.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal.

Terminals, Solder:

1, 2, 3, 4, and 5: Malco No. 2309 or equivalent.

6 and 7: 0.375 x 0.0625 flat copper lug.

8 - Zierick No. 179 or equivalent.

Height: Not to exceed top of case.

Weight: 18.5 ounces maximum.

Altitude: 70,000 feet maximum.

Operating temperature range: -55°C to +170°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 1,000 volts rms.

At reduced barometric pressure: 300 volts rms.

Electrical characteristics:

No load: With 107 volts rms, 400 hertz across (1-2), the current in (1-2) shall not exceed 2 amperes and the power in (1-2) shall not exceed 215 watts.

Voltage across:
(3-5): 49 volts ±10 percent.
(A) (6-7): 5.76 volts ±10 percent.
(7-8): 8.24 volts ±10 percent.

Rated load: With 107 volts rms, 400 hertz across (1-2), the voltage across:
(3-5): 49 volts ct ±10 percent.
(6-7): 5.76 volts ±10 percent.
(7-8): 8.24 volts ±10 percent.



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DC resistance:
(1-2): 0.8 ohm, maximum.
(3-5): 20 ohms, maximum.
(6-7): 0.025 ohm, maximum.
Ⓐ (7-8): 0.25 ohm, maximum.

Polarity: Additive, with 2 and 3, and 5 and 6 connected.

Ⓐ Temperature rise: 99°C with 107 volts rms, 400 hertz across (1-2) at an ambient temperature of 71°C.

Marking location: See figure 1.

Part number: M27/224-01.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0693-04)



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